

1. Write a Pandas program to select the distinct department id from employee's file.

Aim:

To write a Python program using Pandas to display the unique department IDs from the departments dataset.

Algorithm

1. Import the Pandas library.
2. Read the departments.csv file.
3. Select the DEPARTMENT_ID column.
4. Use the unique() function to get distinct department IDs.
5. Display the result.

Code:

```
import pandas as pd
data = pd.read_csv('Csv/Lab1.csv')
distinct_id = data['DEPARTMENT_ID'].unique()
print("Distinct Department IDs:")
print(distinct_id)
```

Output:

```
Distinct Department IDs:
[ 10  20  30  40  50  60  70  80  90 100 110 120 130 140 150 160 170 180
 190 200 210 220 230 240 250 260 270]
```

Result:

Thus, the required operation was performed successfully using the Pandas library.

2. Write a Pandas program to display the IDs of employees who have done two or more jobs in the past.

Aim:

To write a Python program using Pandas to find employees who have worked in two or more jobs. Algorithm

1. Import the Pandas library.
2. Read the job_history.csv file.
3. Group the data by EMPLOYEE_ID.
4. Count the number of job records for each employee.
5. Display employee IDs having two or more job records.

Code:

```
import pandas as pd

data = pd.read_csv('Csv/Lab2.csv')

result = data.groupby("EMPLOYEE_ID").filter(lambda x: len(x) >= 2 ).
sort_values (by="EMPLOYEE_ID")

print("Employees who have done two or more jobs:")

print(result)
```

Output:

Employees who have done two or more jobs:					
	EMPLOYEE_ID	START_DATE	END_DATE	JOB_ID	DEPARTMENT_ID
1	101	1997-09-21	2001-10-27	AC_ACCOUNT	110
2	101	2001-10-28	2005-03-15	AC_MGR	110
7	176	2006-03-24	2006-12-31	SA_REP	80
8	176	2007-01-01	2007-12-31	SA_MAN	80
6	200	1995-09-17	2001-06-17	AD_ASST	90
9	200	2002-07-01	2006-12-31	AC_ACCOUNT	90

Result:

Thus, the required operation was performed successfully using the Pandas library. The desired records were displayed correctly based on the given condition.

3. Write a Pandas program to display the details of jobs in descending sequence on job title.

Aim:

To write a Python program using Pandas to sort the jobs dataset in descending order of job title. Algorithm

1. Import the Pandas library.
2. Read the jobs.csv file.
3. Sort the data using the JOB_TITLE column in descending order.
4. Display the sorted data.

Code:

```
import pandas as pd
data = pd.read_csv('Csv/Lab3.csv')
result = data.sort_values(by="JOB_TITLE", ascending=False)
print(result)
```

Output:

	JOB_ID	JOB_TITLE	MIN_SALARY	MAX_SALARY
11	ST_MAN	Stock Manager	5500	8500
12	ST_CLERK	Stock Clerk	2008	5000
13	SH_CLERK	Shipping Clerk	2500	5500
8	SA_REP	Sales Representative	6000	12008
7	SA_MAN	Sales Manager	10000	20080
9	PU_MAN	Purchasing Manager	8000	15000
10	PU_CLERK	Purchasing Clerk	2500	5500
18	PR_REP	Public Relations Representative	4500	10500
6	AC_ACCOUNT	Public Accountant	4200	9000
14	IT_PROG	Programmer	4000	10000
0	AD_PRES	President	20080	40000
16	MK_REP	Marketing Representative	4000	9000
15	MK_MAN	Marketing Manager	9000	15000
17	HR_REP	Human Resources Representative	4000	9000
13	FI_MGR	Finance Manager	8200	16000
1	AD_VP	Administration Vice President	15000	30000
2	AD_ASST	Administration Assistant	3000	6000
5	AC_MGR	Accounting Manager	8200	16000
4	FI_ACCOUNT	Accountant	4200	9000

Result:

Thus, the required operation was performed successfully using the Pandas library. The desired records were displayed correctly based on the given condition.